

CODERS OF DELHI

Social Network Data Cleaning and Recommendation System

Technology Stack: Python, JSON, Data Structures, Sets, Dictionaries

Project Type: Data Analytics & Recommendation System

Abstract

The Coders of Delhi project is a social network analytics system designed to clean raw user data and generate meaningful recommendations. The project removes duplicate records, inactive users, missing names, and duplicate liked pages while suggesting potential friends and pages.

Project Objectives

- Load social network data.
- Remove duplicate friends.
- Remove inactive users.
- Remove users with missing names.
- Remove duplicate liked pages.
- Generate People You May Know recommendations.
- Generate Pages You Might Like recommendations.

Problem Statement

Real-world social network data often contains inconsistencies and redundant information. These issues reduce data quality and recommendation accuracy. This project addresses these problems through data cleaning and recommendation algorithms.

Methodology

1. Load JSON dataset.
2. Clean data by removing duplicates and invalid records.
3. Analyze friendships for People You May Know.
4. Analyze page preferences for Pages You Might Like.
5. Generate final recommendations.

System Workflow

Raw Data → Load Data → Data Cleaning → Friend Recommendation → Page Recommendation → Final Output

Sample Output

User: Rahul

People You May Know: Aman, Rohit, Priya

Pages You Might Like: Python Community, AI Delhi, Coding Hub

Key Features

- Data Loading
- Data Cleaning
- Duplicate Removal
- Recommendation Engine
- Social Network Analysis

Technologies Used

Python

JSON

Lists, Sets, Dictionaries

Algorithms and Data Structures

Results

The project successfully improves data quality and provides meaningful recommendations while demonstrating practical applications of Python and data analytics concepts.

Future Scope

- Mutual friend scoring
- Popular page ranking
- Graph visualization
- Machine learning recommendations
- Database integration

Skills Demonstrated

- Python Programming
- Data Cleaning
- Data Validation
- Data Structures
- Recommendation Systems

- Problem Solving

Conclusion

The Coders of Delhi project combines data cleaning with recommendation logic to simulate real-world social networking features. It showcases practical programming and analytical skills suitable for academic and portfolio purposes.

Acknowledgement

Thank you to everyone who supported this learning project. It was developed to strengthen practical knowledge of Python, data analytics, and recommendation systems.